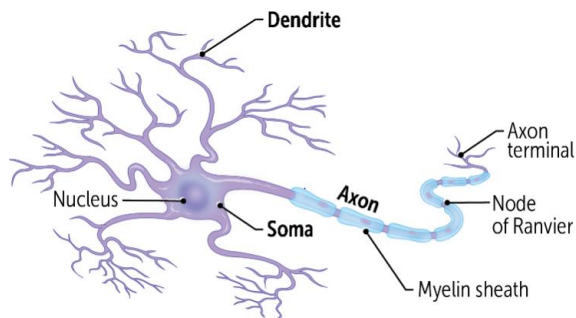




Nervous System

- Formed by neurons and glial cells

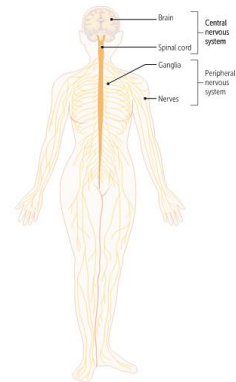
- has two major divisions

1. Central nervous system (CNS)

- consisting of the :
 - **brain and spinal cord**

2. Peripheral nervous system (PNS)

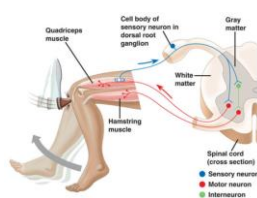
- composed of the:
 - **cranial, spinal, peripheral nerves, and ganglia**



Functions of the Nervous System

1. Sensory function

- **Gathering** information
- **Sensory receptors** detect internal and external stimuli.
- this sensory information is carried into the brain and spinal cord through **cranial** and **spinal** nerves



2. Motor function

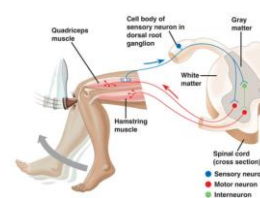
- A **response** to integrated stimuli

- The information is carried out side the brain and spinal cord

- The response activates muscles or glands

3. Integrative function

- **process** and **interpret** sensory input and decide if action is needed



Cont....

Neurons

- Nervous Tissue Consist of 2 types of cells.

1. **Neurons** = nerve cells , the functional and structural unit of the nervous system.

- Specialized to **conduct** information from one part of the body to another.

❖Parts of a Neuron

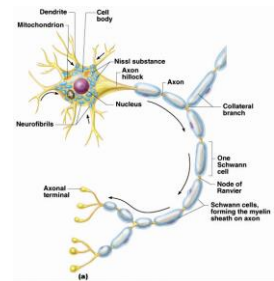
- Most neurons have three parts:

A. Cell body

- contains a **nucleus** surrounded by cytoplasm.
- Biosynthetic center of the neuron

Cont...

- dendrites** and a **single axon** emerge from the cell body.
- Clusters of cell bodies in the CNS and PNS known as **nuclei** and **ganglia** respectively



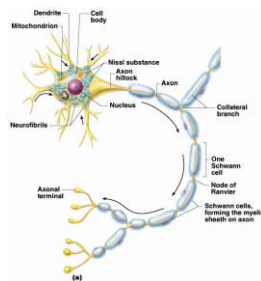
B. Dendrites

- input part of a neuron
- to wards** the cell body

Cont...

C. Axon

- Conducting component, **away** from the cell body
- joins the cell body at the **axon hillock**.
- Some axons have **axon collaterals**
- Ends as axon **terminals**.

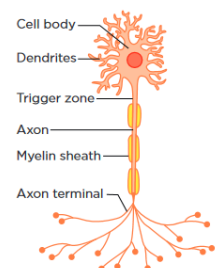


Classification of Neurons

- Structural** and **functional** features are used,

1. Structurally

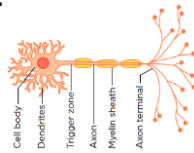
- according to the **number of processes** extending from the cell body.



Cont.

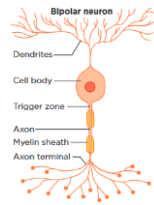
a). Multipolar neurons

- have **several** dendrites and **one** axon.
- E.g. most neurons in the **brain** and **spinal cord**



b). Bipolar neurons

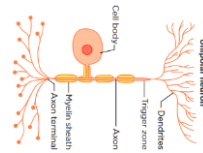
- Have **one main dendrite** and **one axon**.
- E.g. found in **retina** of the eye



Cont...

c. Unipolar neurons

- Have dendrites and one axon that are **fused** together,
- to form a continuous process that emerges from the cell body.
- E.g. Found in **dorsal root ganglia**



Cont...

2. Functionally

- according to the **direction** in which the nerve impulse is conveyed with respect to the **CNS**.

A. Sensory or afferent neurons

- Carry impulses from the sensory receptors **to the CNS**.
- Most sensory neurons are **Unipolar** in structure.

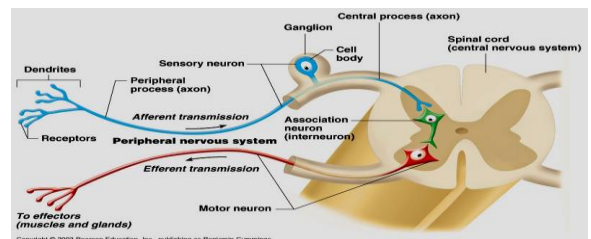
B. Motor or efferent neurons

- convey action potentials **away** from the CNS to **effectors**.
- Most motor neurons are **multi-polar** in structure.

Cont...

C. Interneurons or association neurons

- mainly located within the CNS b/n sensory and motor neurons.
- Most interneurons are **multipolar** in structure.



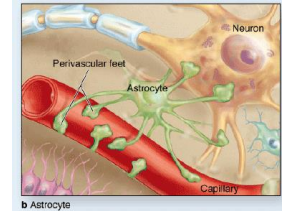
Neuroglia

- Neuroglia are the **supporting** cells of the central nervous system .
- These cells are **non -excitable** and undergo mitotic division.
- Of the six types of neuroglia,
 - **astrocytes**, **oligodendrocytes**, **microglia**, and **ependymal** cells are found only in the **CNS**.
 - **Schwann** cells and **satellite** cells are present in the **PNS**.

Cont...

1. Astrocytes

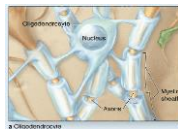
- Star-shaped, abundant, and versatile
- Guide the **migration** of developing neurons.
- Function in nutrient transfer
- **Support** neurons
- **protect** neurons from harmful substances;



Cont...

2. Oligodendrocytes

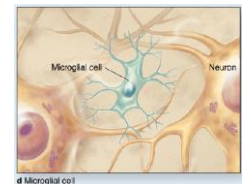
- Produce and maintain myelin sheath around axons of **CNS** neurons
- **insulation** of the axon of a neuron by myelin sheath
- Axons with a myelin sheath are said to be **myelinated**, and those without it are said to be **unmyelinated**
- **White matter** = composed **primarily** of **myelinated** axons.
- **gray matter** = contains neuronal cell bodies, dendrites, unmyelinated axons, axon terminals, and neuroglia



Cont....

3. Microglia

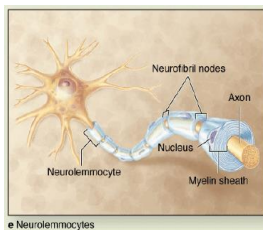
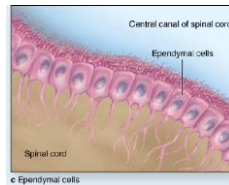
- Specialized immune cells that act as the **macrophages** of the **CNS**
- the main phagocytic cell and **antigen-presenting** cells in the **CNS**



Cont...

4. Ependymal cells

- Line ventricles of the brain and central canal of the spinal cord.
- form cerebrospinal fluid and assist in its circulation



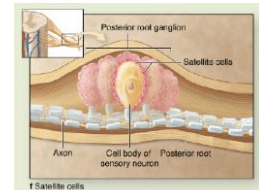
5. Schwann cells

- Produce and maintain myelin sheath around a axon of a PNS neuron.

Cont...

6. Satellite cells

- Support neurons in PNS ganglia
- regulate exchange of materials b/n neurons and interstitial fluid



Central Nervous System

- Consists of:
 - the brain
 - and spinal cord.
- Processes incoming sensory information
- It is also the source of thoughts, emotions, and memories
- Most nerve impulses that stimulate muscles and glands originate in the CNS

- Consists about 100 billion neurons and 10–50 trillion neuroglia

- Have four major parts:

1. brain stem

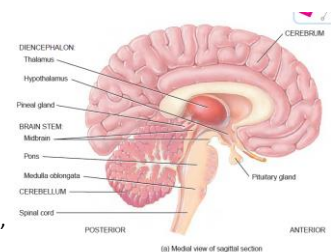
- consists of the medulla oblongata, pons, and midbrain

2. Diencephalon

- Consisting mostly of the thalamus, hypothalamus, and pineal gland.

3. Cerebrum

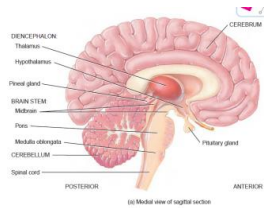
4. cerebellum



Cont....

Cerebrum

- Is the **largest** portion of the brain.
- located in the region of the **telencephalon**.
- It accounts for about **80%**
- is responsible for the **higher mental functions** including **memory** and **reason**.
- consists of the right & left hemispheres connected internally by the **corpus callosum**
- Each hemisphere contains a central cavity called the **lateral ventricle**.



Cont...

- The cerebrum consists of two layers.

A. Cerebral cortex

B. White matter Beneath the cerebral cortex and constitutes the second layer.

❖ **The Cerebral cortex forms the surface layer**

- is composed of gray matter(nerve cell bodies.)
- has numerous folds and grooves called **convolutions**.
- The elevated folds of the convolutions are the cerebral **gyri** (singular, **gyrus**)
- the grooves are the cerebral **sulci** (singular, **sulcus**)

Cont....

- The cerebrum consists of two layers.

A. Cerebral cortex

B. White matter Beneath the cerebral cortex and constitutes the second layer.

❖ **The Cerebral cortex forms the surface layer**

- is composed of gray matter(nerve cell bodies.)
- has numerous folds and grooves called **convolutions**
- The elevated folds of the convolutions are the cerebral **gyri** (singular, **gyrus**)
- the grooves are the cerebral **sulci** (singular, **sulcus**)

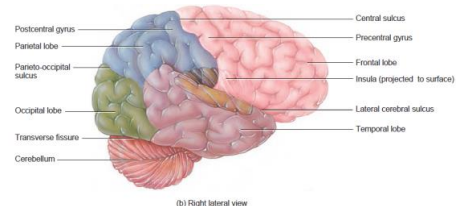
Cont...

❖ **Main sulci:**

- central, lateral, parieto -occipital & calcarine sulci

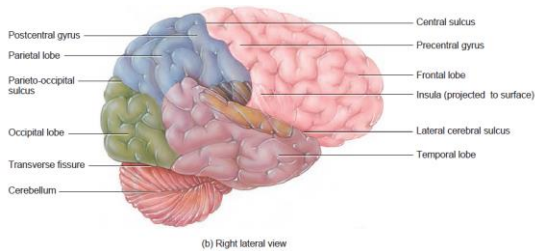
❖ **Main gyri:**

- **precentral gyrus:** anterior to central sulcus
- **postcentral gyrus:** post.to central sulcus
- **Supramarginal gyrus:** around end of lateral sulcus.



Cont...

- **Angular gyrus:** around end of sup.temporal fissure
- **Superior temporal:** b/n lat. & sup.temporal fissure
- **Cingulate gyrus:** b/n cingulate sulcus & corpus callosum

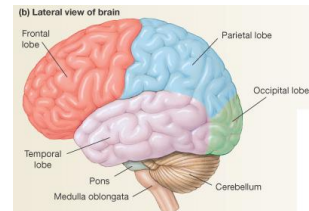


Cont...

- Each cerebral hemisphere has four lobes that are named after the bones that cover them

❖ **Frontal Lobe:**

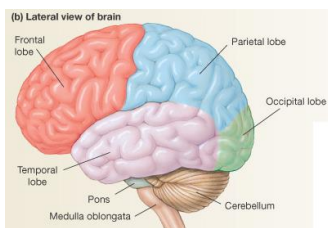
- forms the anterior portion of each hemisphere
- extends from the **central** sulcus to the **frontal** pole
- lies above the lateral sulcus and anterior to the central sulcus.



Cont...

❖ **Parietal lobe:**

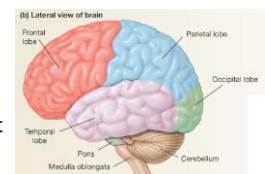
- extends from the **central** sulcus to the occipital lobe
- lies superior to the temporal lobe,
- posterior to the central sulcus.
- **functions** : receives and integrates **sensory** information



Cont....

❖ **Temporal lobe:**

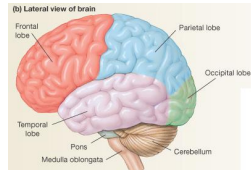
- lying below the **lateral** sulcus
- located below the parietal lobe and the posterior portion of the frontal lobe
- is separated from both by the **lateral sulcus**
- contains **auditory** centers that receive sensory fibers from the cochlea of the ear



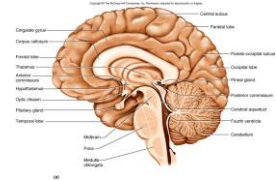
Cont....

❖ **Occipital lobe:**

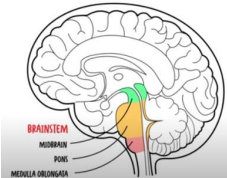
- lies posterior to the **parieto-occipital sulcus**
- forms the posterior portion of the cerebrum
- lies superior to the cerebellum
- contains the **visual cortex**
- The principal function of the occipital lobe is concerned with **vision**.

**Diencephalon**

- is the second subdivision of the forebrain and divide in to:
 1. Thalamus
 2. hypothalamus
 3. the epithalamus
- The 3rd ventricle forms a cavity on the median plane within the diencephalon

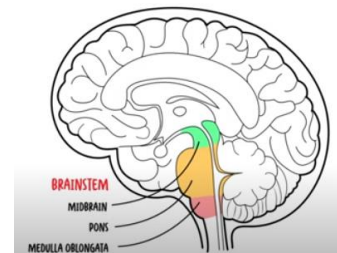
**Brain Stem**

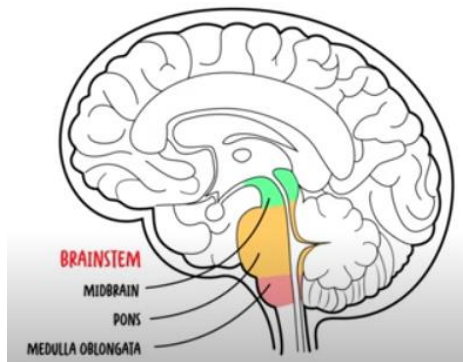
- b/n the **spinal cord** and the diencephalon
- regulates **heart rate** and **respiration**
- Allows communication between the cerebrum, cerebellum and spinal cord
- Contains the **cardiovascular, respiratory, vomiting** and **vasomotor** centers
- Is passageway for many **neural pathways**



Cont...

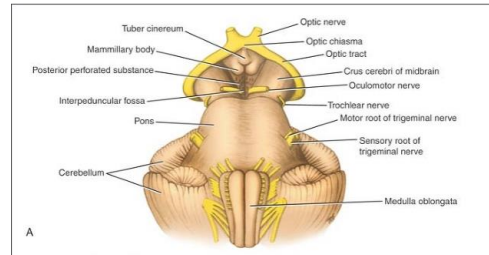
- It consists of three regions:
 - I. medulla oblongata
 - II. Pons
 - III. midbrain





Midbrain

- connects the **pons** to the diencephalon
- traversed by the **cerebral aqueduct**
- contains the nuclei of **Oculomotor(III)** and **Trochlear (IV)** nerves



Pons

- is above the **medulla** and anterior to the **cerebellum**.
- connects the **medulla oblongata** to the **midbrain**.
- contains the nuclei of **trigeminal (V)**, **abducens (VI)**, and **facial (VII)**.

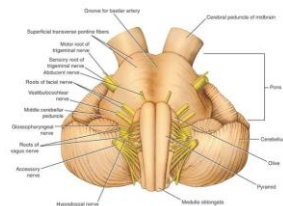
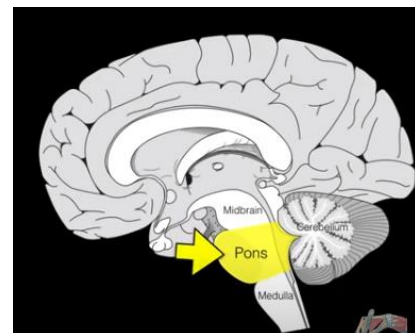
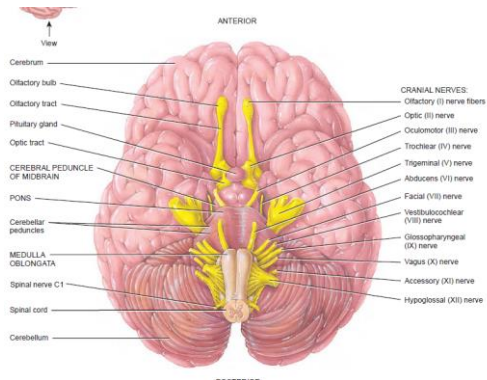


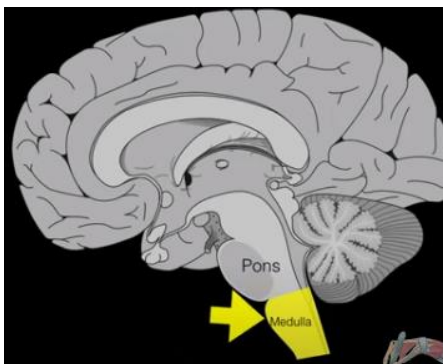
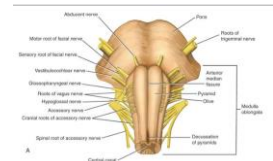
Figure 9-17 Anterior surface of the brainstem showing the pons.





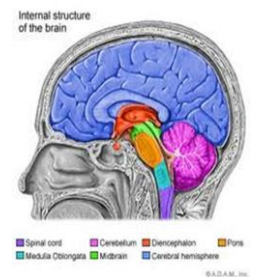
Medulla Oblongata

- connects the **pons** superiorly with the **spinal cord** inferiorly.
- contains the nuclei of vestibulocochlear (VIII), glossopharyngeal (IX), vagus (X), accessory (XI) and hypoglossal (XII) nerves.



Cerebellum

- is the 2nd largest structure in the brain.
- Located
 - Posterior to the medulla and pons
 - Below the cerebrum
- has two lateral cerebellar hemispheres joined by **vermis**
- attached to the brain stem by **inferior, middle & superior cerebellar peduncles**



External features of cerebellum

➤ It has two notches

1. anterior notch

- is the wider and lodging the back of pons & medulla
- It is separated from them by cavity of 4th ventricle



2. posterior notch

- occupied by falx cerebelli, which separates the 2 cerebellar H

2. Superior surface

- lies beneath tentorium cerebelli
- Has:
 - a raised superior vermis
 - a large cerebellar hemisphere on each side
 - primary & horizontal fissures



Cont...

➤ It has two surfaces

1. Inferior surface

- rounded on each side and presents :
 - a deep groove (vallecula) b/n the 2-cerebellar hemispheres, which is occupied by the inferior vermis
- ✓ Tonsil is a small part of cerebellar hemisphere that lies lateral to inferior vermis



Cont...

➤ It has the fissures and folia

- Three main fissures

1. Primary fissure

- V-shaped, well defined fissure, lies on superior surface
- separates the small anterior lobe from the larger middle lobe (or posterior lobe)



Cont...

Cont...

2. Horizontal fissure

- lies along the **sides** of cerebellum, extending from anterior notch to posterior notch
- separates the **superior** from the **inferior** surfaces of the cerebellum



Cont....

3. Secondary (posterolateral) fissure

- lies on **inferior** surface of the cerebellum
- separates **flocculo-nodular** lobe from the **remainder** of cerebellum



❖Function

- It regulates posture and balance
 - **Coordinating** skeletal muscle contractions by recruiting precise motor units within muscles.
 - Impulses for voluntary muscular movement originate in the cerebral cortex and are **coordinated** by the cerebellum
- The cerebellum can be permanently damaged by **trauma** or stroke or temporarily affected by drugs such as alcohol.
- These alterations can produce **ataxia**, a disturbance in balance.

Meninges

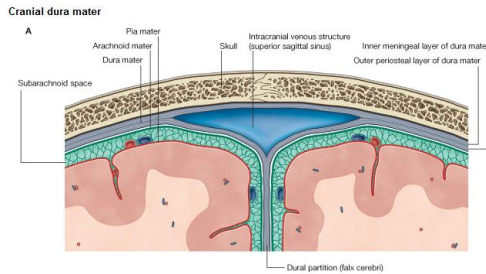
- Are membranous coverings of the brain and spinal cord.
- Cranial Meninges lie immediately internal to the **cranium**.
- Protect the brain and spinal cord.
- Form the supporting framework for arteries, veins, and venous sinuses.
- **Enclose** a fluid-filled cavity, the **subarachnoid** space, which is vital to the normal function of the brain.

Cont...

- are composed of three membranous connective tissue layers

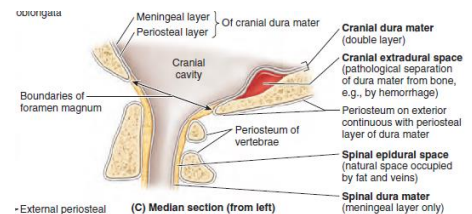
1. **Dura mater**: tough, thick external fibrous layer.
2. **Arachnoid mater**: thin intermediate layer.
3. **Piamater**: delicate internal vasculated layer.

- *Leptomeninges* make up of the **arachnoid** and **pia mater**.
- The arachnoid is separated from the pia by the **subarachnoid space**, which contains cerebrospinal fluid (CSF).



❖ Dura Mater

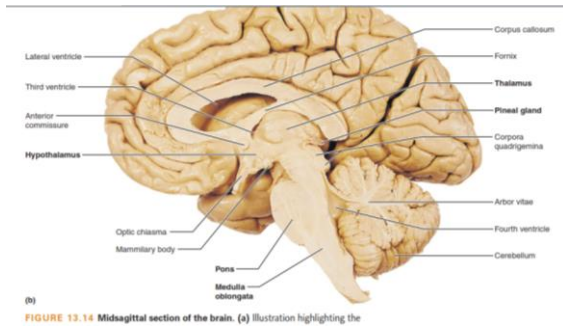
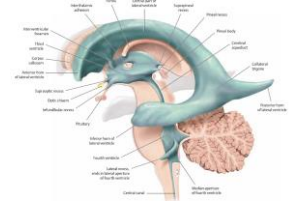
- Consists of an outer **periosteal** layer and an inner **meningeal** layer.
- The **external periosteal** layer is continuous at the cranial foramina with the **periosteum** on the external surface of the **calvaria**.
- And is **not** continuous with the dura mater of the **spinal cord**.
- the **inner meningeal** layer is continuous with the **spinal dura** mater through the foramen magnum.



Cont...

1. Lateral ventricle

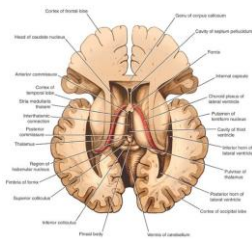
- two large ventricles, and one is present in each cerebral hemisphere.
- Roughly C-shaped cavity
- Connected through **interventricular foramen** to 3rd ventricle.



Cont...

2. Third ventricle

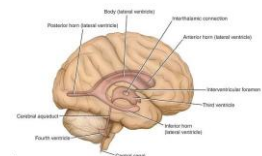
- found in the diencephalons, b/n 2 **thalami**.
- **cerebral aqueduct** connects it to the fourth ventricle



Cont...

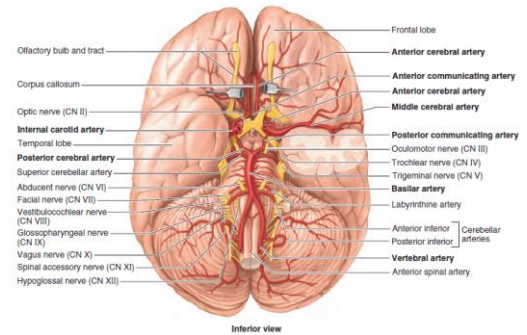
3. Fourth ventricle

- found at the base of **cerebellum**
- On the posterior surface of medulla **oblongata** and **pons**
- Anterior to the cerebellum.
- It continues with the **central canal** of the spinal cord



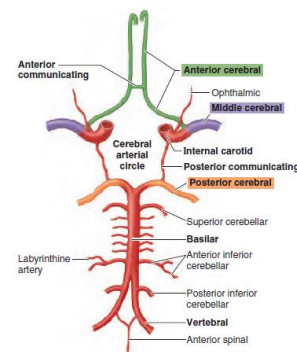
Arterial Blood Supply to Brain

- is derived from the **internal carotid** and **vertebral** arteries.
- The **terminal branches** of the **internal carotid** arteries are:
 - the **anterior** and **middle cerebral** arteries.
- **Vertebral artery**.:
 - Basilar a.
 - Posterior cerebral a.
 - Anterior Spinal a.



Cont...

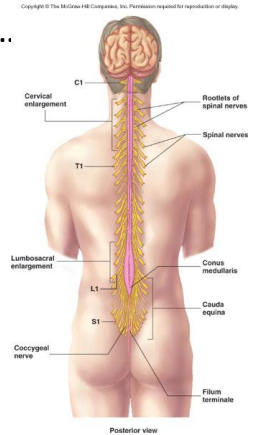
- The **cerebral** arterial circle (of **Willis**):
 - is an arterial **anastomosis** b/n arteries supplying the brain,
- Formed by the:
 - Anterior communicating artery
 - Anterior cerebral artery
 - Internal carotid arteries
 - Posterior communicating arteries
 - Posterior cerebral arteries
- ❖ **Venous drainage** from the brain occurs via **cerebral** and **cerebellar veins** that drain to the adjacent **dural venous sinuses**.



Spinal cord

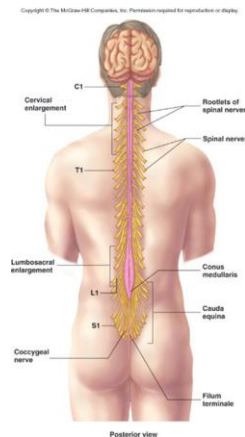
Cont...

- Extends from **foramen magnum** to **L2** in adults
- in newborns, it extends to L3.
- continuous with the medulla oblongata at the **spinomedullary** junction.
- Consists of:
 - Cervical region
 - Thoracic region
 - Lumbar region
 - Sacral region
 - Coccygeal region



Cont...

- Its terminal Part is Called **Conus Medullaris**.
- The **filum terminale**, a fibrous strand composed of **pia mater**, extends inferiorly from the conus medullaris to the **coccyx**

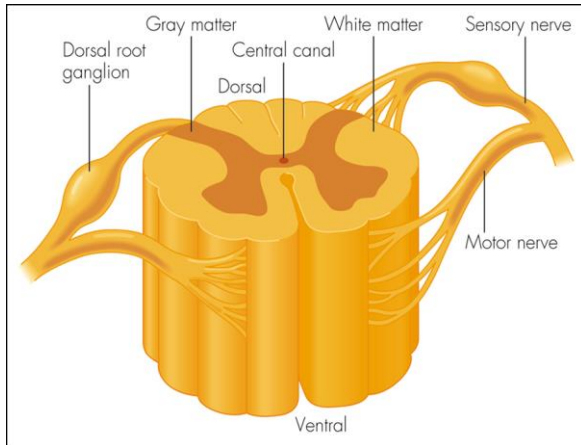


Internal Morphology

- In transverse sections, the spinal cord consists of central gray matter and peripheral white matter.

A. Gray matter

- is located centrally within the spinal cord
- contains neuronal **cell bodies**, **dendrites**, **unmyelinated axons**, axon terminals, and **neuroglia**.
- is butterfly- or H- shaped in a configuration that varies according to spinal cord level.
- contains a **central canal**.
- is divided into **three horns** or **cell columns** on each side named relative to their location.



Cont...

1. Posterior horn

- Contain cell bodies and axons of **interneurons** as well as axons of **incoming sensory** neurons.
- Contains **somatic** and **visceral** sensory nuclei

2. Lateral horn

- receives **viscerosensory** input.
- is found b/n the dorsal and ventral horns.
- present only in **thoracic** and **upper lumbar** segments of the spinal cord (T1-L2).
- contain cell bodies of **autonomic motor** neurons

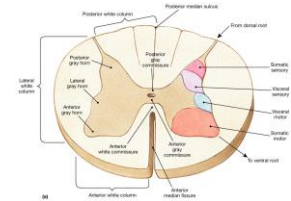
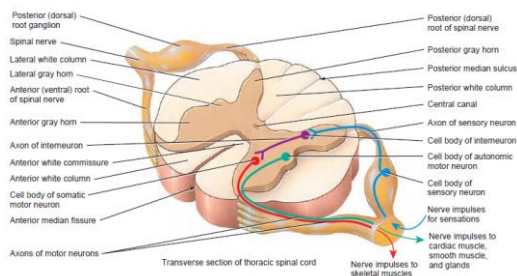
3. Anterior horn

- contain cell bodies of **somatic motor** neurons

Cont...

B. white matter

- consists primarily of **myelinated** axons of neurons
- organized into **anterior**, **lateral**, and **posterior white columns**
- Each column contains one or more **tracts**

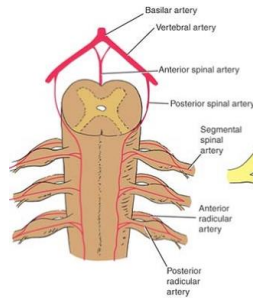


Blood Supply of the Spinal Cord

- Receives its arterial supply from:
- two **posterior** spinal arteries
- and the **anterior** spinal artery.

➤ **Segmental Spinal Arteries**

- reinforcing arteries
- branches of arteries outside the vertebral column.



Peripheral nervous system

Cont...

- is that portion of the nervous system outside the CNS.
- The PNS consists of **sensory** and **motor** neurons.
- The sensory neurons **collect** information and convey to the CNS
- motor neurons **convey** information from CNS to muscle and glands
- The nerves of the PNS are classified as
 - **Cranial nerves** (I-XII) pairs and
 - **Spinal nerve** (31) pairs

cont...

PNS -subdivided into

I. **Somatic nervous system**

Consist of :

- **Sensory neurons**- convey information from **somatic** receptors and **special** senses to **CNS**
- **Motor neurons** –conduct impulses from CNS to skeletal muscle only- voluntarily

Cont..

II. The Autonomic Nervous System (ANS) Consist of:

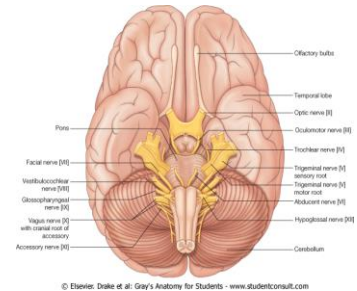
- **Sensory neurons**- convey information from autonomic sensory receptors, located primarily in **visceral** organs to **CNS**
- **Motor neurons**- conduct impulse from CNS to **smooth** muscle, **cardiac** muscle and glands- involuntary

Cranial Nerves

• **12 pairs of cranial nerves.**

• **All are distributed in the head & neck, except CN-X.**

• **All nerves except CN-XI originate from the brain.**

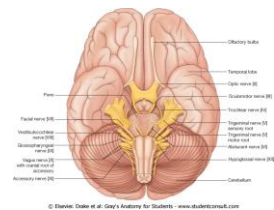


© Elsevier, Drake et al: Gray's Anatomy for Students - www.studentconsult.com

Cont....

- The 12 pairs of cranial nerves can be grouped - according to their **functional neuronal** components:
- ✓ **Sensory** nerves: carry axons of **sensory** neurons.
- ✓ **Motor** nerves: contain only axons of **motor** neurons as they leave the brain stem.
- ✓ **mixed** nerves: contain axons of **both** **sensory** and **motor** neurons.

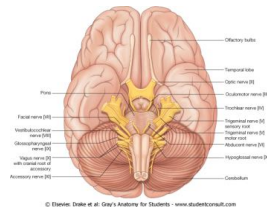
- I **Olfactory nerve** – sensory for smell
- II **Optic nerve** – sensory for vision
- III **Oculomotor nerve** – motor fibers to eye muscles
- IV **Trochlear** – motor fiber to eye muscles
- V **Trigeminal nerve** – sensory for the face; motor fibers to chewing muscles



© Elsevier, Drake et al: Gray's Anatomy for Students - www.studentconsult.com

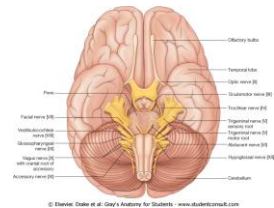
Cont....

- VI **Abducens nerve** – motor fibers to eye muscles
- VII **Facial nerve** – sensory for taste; motor fibers to the face
- VIII **Vestibulocochlear nerve** – sensory for balance and hearing
- IX **Glossopharyngeal nerve** – sensory for taste; motor fibers to the pharynx



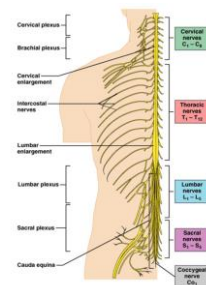
Cont....

- X **Vagus nerves** – sensory and motor fibers for pharynx, larynx, and viscera
- XI **Accessory nerve** – motor fibers to neck and upper back
- XII **Hypoglossal nerve** – motor fibers to tongue



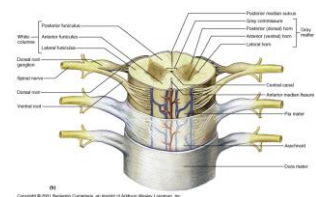
Spinal Nerves

- 31 pairs
- Connect to the **spinal cord**
- Named for point of tissue from the spinal cord
 - 8 pairs of cervical nerves (C₁-C₈)
 - 12 pairs of thoracic nerves (T₁-T₁₂)
 - 5 pairs of lumbar nerves (L₁-L₅)
 - 5 pairs of sacral nerves (S₁-S₅)
 - 1 pair of coccygeal nerves (Co₁)
- formed as the fibers from the posterior and anterior roots converge and emerge through an **intervertebral foramen**



at its origin the spinal nerve has two roots:

1. **Anterior (ventral) root**
 - composed of **motor** fibers
 - the axons of nerve cells in the **anterior** horn and **lateral** horn.
2. **Posterior (dorsal) root**
 - composed of **sensory** fibers
 - has swelling called posterior (dorsal) root ganglia, where the cell bodies of sensory neurons are located

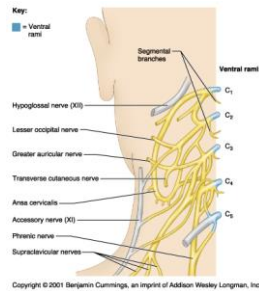


Nerve plexuses

- are networks of nerves formed from ventral rami
- except in the thoracic nerves T2-T12, the ventral rami of spinal nerves combine and then split again as networks of nerves referred to as **plexus**

1. Cervical plexus (anterior rami C1-C4)

- lies **anteromedial** to the levator scapulae and middle scalene muscles and deep to the SCM
- cutaneous** (sensory) branches and **motor** branches



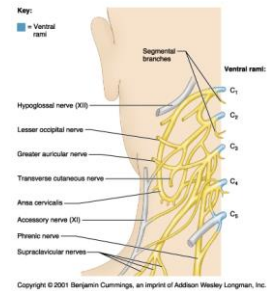
Branches:

■ Cutaneous branch

- lesser occipital (C2,C3)
- greater auricular (C2,C3)
- transverse Cutaneous nerve (C2,C3)
- supra clavicular (C3&C4)

■ Motor branches

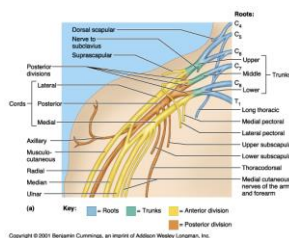
- phrenic nerve (C3, C4, and C5), motor supply to diaphragm
- branch to infrahyoid muscle (hypoglossal nerve -Ansa cervicalis)



Cont....

2.Brachial plexus (C5-T1)

- found partly in the **neck** and partly in the **axilla**
- innervates **upper limb**, **shoulder** and **neck** muscles
- structurally divided in to:
 - roots
 - trunk
 - divisions
 - and cords

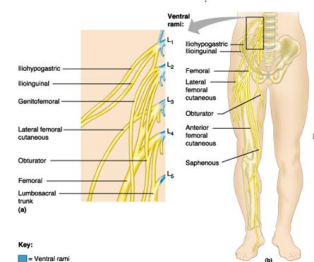


3. Lumbar plexus

- Formed anterior to the lumbar transverse processes, within the proximal attachment of the **psoas** major
 - anterior rami of **L1** through **L4** nerves
- branches of the lumbar plexus:

1. Femoral nerve(L2-L4)

- Muscular branch**- quadriceps femoris, sartorius, pectineus and iliacus

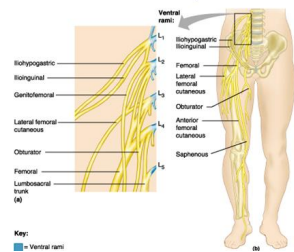


Cont....

- **Cutaneous branch**- skin of the anterior and medial thigh and skin of the medial leg and foot.

2. Obturator nerve(L2-L4)

- **Muscular branch**- supply adductor muscles
- **Cutaneous branch**- skin of medial thigh



3. lumbosacral trunk (L4, L5)

- participate in the formation of the sacral plexus.

4. ilioinguinal and iliohypogastric nerves (L1)

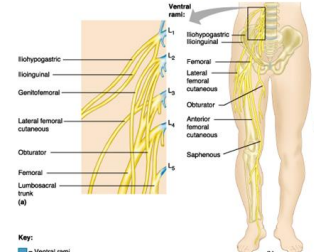
- supply the abdominal muscles and skin of the inguinal and pubic regions

5. The genitofemoral nerve (L1, L2)

6. lateral cutaneous nerve of the thigh(L2, L3),

- supplies skin on the anterolateral surface of the thigh

Cont....



4. Sacral plexus

- located on the posterolateral wall of the lesser pelvis

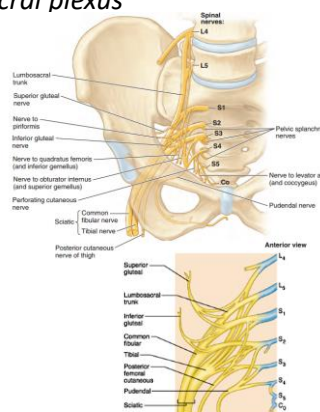
Nerves from the sacral plexus

1. Superior gluteal nerve

- L4, L5, S1
- gluteus medius, minimus

2. Inferior gluteal

- L5, S1, S2
- gluteus maximus

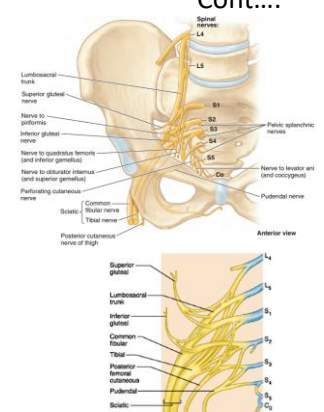


3. Pudendal nerve

- S2, S3, S4
- supplies most of the skin and muscles of perineum and external sphincters

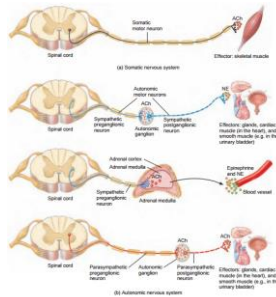
4. Sciatic nerve

- L4, L5, S1, S2, S3
- the thickest and longest nerve of our body
- it branches in to two:
 - Tibial
 - Common peroneal nerve



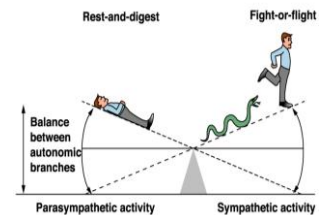
Cont....

- two neurons span the distance from CNS to effector
- the first begins with a soma in the CNS and terminates in a ganglion of the PNS it's called **preganglionic (presynaptic)** neuron
- in the ganglion it synapse with a **post ganglionic (post synaptic)** neuron whose axon extends the rest of the way to the target organs



Cont....

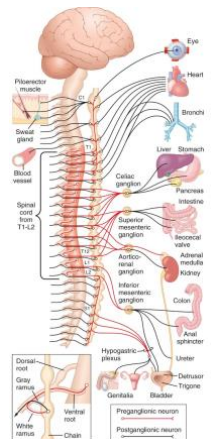
- divided into:
- **Sympathetic Nervous System**
- "Fight or Flight"
- **Parasympathetic Nervous System**
- "Rest and Digest"



98

I. Sympathetic

- Also called the **thoracolumbar** division of ANS
- Preganglionic neurons exit **T1-L2** (spinal nerves) and lead to sympathetic chain of ganglia (**Para vertebral ganglia**) along each side of the vertebral column
- the preganglionic neurons are **myelinated** and post ganglionic neurons are **unmyelinated**



II. Parasympathetic

- known as the **craniosacral**
- preganglionic neuron originate in the **brain** (CN III, VII, IX & X) and **S2-S4**
- preganglionic neuron synapse with post ganglionic neuron **near or with in** effector organ (terminal or intramural ganglia)

